

## **UAS SafeNet ProtectServer HSM Deployment Guide - 6.0.1-GA (AI)**

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# 1. Preface - UAS SafeNet ProtectServer HSM Deployment Guide

This guide explains how to deploy, configure, and setup SafeNet ProtectServer Network HSM (Hardware Security Module) to manage and store digital keys and messages used during transactions, for identification, and applications access securely. It protects the cryptographic infrastructure by provisioning encryption, decryption, authentication, and digital signing services.

## Intended Reader

This document is intended as a reference for the administrators and the security officers.

## Document Scope

This document shows you how to:

- Install and setup SafeNet ProtectServer Network HSM with AccessMatrix UAS.
- Customise from the SafeNet documentation the required information and configurations for use with AccessMatrix UAS, the FM (Functionality Module), and the backup mediums such as the smartcards and others.



**Note:** Certain sections in this guide direct you to the related AccessMatrix and SafeNet documentation to avoid duplication. You can refer to SafeNet documentation for the information you need to install and deploy SafeNet ProtectServer Network HSM.

For any feedback or questions regarding this document, contact [doc@i-sprint.com](mailto:doc@i-sprint.com).

## Related Documents

For more information, see the following documents in the AccessMatrix UAS (Universal Authentication Server) documentation set:

- [\*OneSpan FM with SafeNet ProtectServer HSM Implementation Guide\*](#)
- [\*AccessMatrix Common Administration Guide\*](#)
- [\*UAS Administration Guide\*](#)

For SafeNet documentation, refer to:

- *SafeNet: Network HSM Installation and Configuration Guide*
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- *SafeNet: PTK-C Administration Guide*
  - *SafeNet: PSESH Command Reference Guide*

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## 2. Introduction to UAS SafeNet ProtectServer HSM Deployment

SafeNet ProtectServer Network HSM is a security-hardened network crypto device that provides a wide range of cryptographic functions that include high-speed encryption, user and data authentication, message integrity, as well as secure key storage and key management.

### System Requirements

The following hardware and software are required for the deployment of SafeNet ProtectServer Network HSM.

|          |                                                                                                                                                       |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>i</b> | <b>Note:</b> SafeNet ProtectServer External 2 (PSE2) HSM is an older PSE 2 HSM which has already been rebranded to SafeNet ProtectServer Network HSM. |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Hardware

- Host or AccessMatrix Server
- SafeNet ProtectServer Network Hardware Security Module (HSM)

|          |                                                                                                                                                                                       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>i</b> | <b>Note:</b> AccessMatrix server is optional for deployment if you are using a SafeNet ProtectServer Network HSM without a host. You can connect a monitor and a keyboard to the HSM. |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Software

- SafeNet ProtectServer HSM drivers - Protect Toolkit C version 5.7.0 with firmware to be used for communications:
  - SafeNet Network HSM Access Provider 5.7.0
  - SafeNet ProtectToolkit C Runtime 5.7.0
  - SafeNet ProtectToolkit C SDK 5.7.0

|          |                                                                                                                                                                                                                                                                                                                                                              |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>i</b> | <b>Notes:</b> <ul style="list-style-type: none"><li>• Supported environment: Oracle JRE/JDK 1.8 (minimum is 1.8.0_181_b13).</li><li>• Supported platforms: Windows, Linux, AIX, HP-UX and Solaris.</li><li>• SafeNet ProtectServer HSM drivers, Protect Toolkit C version 5.7.0 provide SDKs for Windows, Linux, AIX, HP-UX and Solaris platforms.</li></ul> |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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### 3. Configure SafeNet ProtectServer Network HSM with AccessMatrix UAS

These following sections provide details on configuring the SafeNet ProtectServer Network HSM:

- [Deploy SafeNet ProtectServer Network HSM](#)
- [Initialize Token and Change Pin \(Windows OS\)](#)
- [Manage Keys](#)
- [Back up the Selected Keys to Smartcards](#)
- [Import the Selected Keys from Smartcards](#)
- [Configure Secure Messaging System \(SMS\)](#)
- [Setup Functionality Module \(FM\) in HSM](#)
- [Manage System Events Log](#)
- [Configure HSM](#)
- [Update and Tamper Firmware](#)

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## 3.1. Deploy SafeNet ProtectServer Network HSM

The following procedural steps show you how to deploy a SafeNet ProtectServer Network HSM with the AccessMatrix server:

| Item | Tasks                                                                                             |
|------|---------------------------------------------------------------------------------------------------|
| 1.   | <a href="#">Install SafeNet ProtectServer Network HSM</a>                                         |
| 2.   | <a href="#">Configure SafeNet ProtectServer Network HSM IP Address</a>                            |
| 3.   | <a href="#">Test Network Access from AccessMatrix Server to SafeNet ProtectServer Network HSM</a> |
| 4.   | <a href="#">Install Oracle JDK</a>                                                                |
| 5.   | <a href="#">Install SafeNet ProtectServer Network Client drivers</a>                              |
| 6.   | <a href="#">Test the Connection to SafeNet ProtectSever Network HSM</a>                           |

### Install SafeNet ProtectServer Network HSM

Refer to SafeNet document **Network\_HSM\_Installation\_and\_Configuration\_Guide.pdf** section *Installing the SafeNet ProtectServer Network HSM Hardware* to install HSM hardware. The instructions show you how to:

1. Connect HSM to the power supply.
2. Connect HSM to the network.
3. Turn ON the HSM and notice its display shows LINUX OS bootup.
4. After the installation, test if the HSM is configured properly. At the command line, `ping` to the HSM to check its connection from AccessMatrix server.
5. The default passwords for the users are as follows if this is the first time you log in:

| User name   | Default password |
|-------------|------------------|
| admin       | password         |
| audit       | password         |
| pseoperator | password         |

- a. Login as admin and enter the admin password to log in to HSM.
-

- 
- b. To test if HSM is set up and running correctly, enter the diagnostic utility `hsmstate` in the command line.

The response below shows that the HSM is functioning normally.

```
HSM device 0: HSM in NORMAL MODE. RESPONDING. Usage Level=0%
```

6. Alternatively, on the AccessMatrix server, you can access the Secure Shell (SSH) network by downloading the SSH client such as PuTTY from the [www.putty.org](http://www.putty.org) website.
  - a. 6.1 Evoke the `Putty` program. Enter the Host Name (or IP address) and select **Open** on the PuTTY **Configuration** dialog box.
  - b. 6.2 Login as admin, and then enter the admin password.
  - c. 6.3 Type the command `status` in the psesh (it denotes PSE shell) command line to see a list of status commands you can use.
  - d. 6.4 Type `status ma` to test if the HSM is functioning normally.

The response below shows the HSM is functioning normally:

```
eth0 00:01:4E:02:D0:73
eth1 00:01:4E:02:D0:74
Command Result: 0 (success)
```

**i Note:** As a security measure, the admin account is locked out after 10 consecutive failed login attempts using the console (serial port or keyboard and monitor). Further login attempts will produce a message like the following: "Your admin account is locked due to 11 failed logins." You need to tamper the HSM and reboot the system to reset the admin password. Refer to [Tamper Firmware](#) for more information.

## Configure SafeNet ProtectServer Network HSM IP Address

1. Next, set up the network IP address of the HSM.

There are two NIC ports (eth0 and eth1) on the SafeNet ProtectServer Network HSM.

- a. In the psesh command line, type `help` and press Enter to list the available commands.
- b. Type `network` and press Enter to see a list of subcommands.
- c. Type `network in` to view the syntax of how you can edit the IP address for the network device.
- d. Follow the syntax below to change the IP address.

Syntax: `network interface -device <netdevice> ip <ipaddress> -netmask <ipaddress> [-gateway <ipaddress>] [-force]`

For example: `[PSE-II] psesh:> network interface -device eth0 -ip`

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```
192.168.1.223 -n 255.255.255.0
```

You must restart the network service for the new network settings to take effect. During restart, it displays:

```
If you are sure that you wish to restart the network, type 'proceed',  
otherwise type 'quit'.  
> type proceed  
Proceeding...  
Restarting network service...  
Shutting down interface eth1: [OK]  
Shutting down loopback interface: [OK]  
Bringing up loopback interface: [OK]  
Bringing up interface eth0: Determining if ip address 192.168.1.223 is  
already in use for device eth0... [OK]  
Bringing up interface eth1: Determining if ip address 192.168.1.222 is  
already in use for device eth1.. [OK]  
Command Result: 0 (Success)
```

2. Type `network show` in the [PSE-II] psh command line to view the current network settings.
3. To edit the hostname, for example, type `network h PSE-network` in the command line. The response below shows that a new hostname is created successfully.

```
Success: Hostname PSE-network set.
```

Notice the command line changes to [PSE-network] psh.

4. To set up the domain name, for example, type `network do SafeNetPSENWHSM` in the command line and press Enter. The response below shows that the domain name is created successfully.

```
Success: DomainName SafeNetPSENWHSM set.
```

5. Type `network s` to see the changed network domain and hostname.

## Test Network Access from AccessMatrix Server to SafeNet ProtectServer Network HSM

Use one of the methods below to test the connection.

### **Method 1:**

Ping the HSM IP address from AccessMatrix server or the client hosting server/computer.

### **Method 2:**

Use the PuTTY tool to log in to the SafeNet HSM (SSH server). Enter the hostname (IP address), Port and the connection type - SSH to open it.

Log in as admin, and type the admin password.

---

**i Note:** The users, admin, audit and pseoperator can access the HSM. See SafeNet: PSESH\_Command\_Reference\_Guide on the topic "User" for more information.

## Install Oracle JDK

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Windows | If AccessMatrix server or the client hosting server/computer is running on Window OS, use the procedural steps below to install Oracle JDK.                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>1. Download and install the JDK executable from Oracle website.</div> <div>2. Set an environment variable with the following details:</div>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>Variable name: JAVA_HOME</div> <div>Variable value: &lt;JDK_Installation_Folder&gt; (e.g. C:\Program Files\Java\jdk1.8.0_181)</div>                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>3. You will also be required to add bin location of your java binaries to the systems PATH variable. Edit PATH variable under System variables and append string "%JAVA_HOME%\bin" in variable value and save it.</div>                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>i</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <div><b>Note:</b> Java environment is required to execute the batch files below after you install the SafeNet ProtectServer network client drivers:</div> <div><ul style="list-style-type: none"><li>• <b>gCTAdmin.bat</b> (This is a utility to initialize the token and adapters for SafeNet HSM.)</li><li>• <b>KMU.bat</b> (This is the Key Management Utility application required to manage the secret key, public key, transport key and more for SafeNet ProtectServer Network HSM.)</li></ul></div> |
| Linux   | If AccessMatrix server or the client hosting server/computer is running on Linux OS, use the procedural steps below to install Oracle JDK.                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>1. Download and install the JDK rpm file from Oracle website.</div> <div><code>rpm -ivh &lt;jdk_package&gt;.rpm</code></div> <div>2. Add the Java location path in JAVA_HOME. For example:</div>                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div><code>export JAVA_HOME=&lt;jdk or jre installation path&gt;</code></div> <div><code>export PATH=\$JAVA_HOME/bin:\$PATH</code></div>                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|         | <div>i</div> <div><b>Note:</b> Java environment is required to execute the batch files below after installing SafeNet ProtectServer Network client drivers on Linux OS:</div> <div><ul style="list-style-type: none"><li>• <b>gCTAdmin.bat</b> (This is a utility to initialize the token and adapters for SafeNet HSM.)</li><li>• <b>KMU.bat</b> (This is the Key Management Utility application required to manage the secret key, public key, transport key and more for SafeNet ProtectServer Network HSM.)</li></ul></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Install SafeNet ProtectServer Network Client Drivers

Refer to the SafeNet ProtectToolkit 5.7 Product Documentation *Network HSM Installation/Configuration Guide* to connect SafeNet ProtectServer Network HSM to a client machine or AccessMatrix server in the network mode. It shows you how to assign an IP address, hostname, gateway and access control. Next, you retrieve the client drivers from SafeNet ProtectServer HSM package version 5.7.0.



### Notes:

- Make sure you obtain the Runtime package for operational use and access to an HSM and not the SDK package for developer use.
- Before the Network client drivers' installation, make sure the .NET Framework version 4.5 or later version and Microsoft Visual C++ (MSVC) 2010 redistributable runtime packages are installed for building, deploying, and running applications and services that use .NET technologies, such as desktop applications and Web services.
- AccessMatrix server is already set up and installed with its application and licenses.

### **a. Install Network HSM Access Provider**

This package, HSM Access Provider must be installed to enable connection of AccessMatrix server to one or multiple HSMs. It acts as a data abstraction layer for which you can add multiple front-end Application Program Interfaces (APIs); they are high-level cryptographic APIs in a network mode.

1. Run the **PTKnethsm.msi** file in the folder *network\_hsm\_access\_provider* and follow the instructions as displayed on the wizard dialog box to complete the installation.
2. Enter the IP Address or name of the SafeNet ProtectServer HSM when prompted during the installation. Notice the server configuration string is stored in the Windows registry as a configuration item (ET\_HSM\_NETCLIENT\_SERVERLIST).
3. Click **Close** to exit the installation wizard. The driver's binary directory is automatically appended to PATH environment variable.
4. After the installation, run the `hsmstate` command in the command prompt to verify whether HSM Access Provider is installed successfully. The response below shows that the HSM is functioning normally.

```
HSM device 0: HSM in NORMAL MODE. RESPONDING. Usage Level=0%
```



**Note:** The HSM IP Address configuration is stored in HKEY\_LOCAL\_MACHINE\SOFTWARE\Safenet\HSM\NETCLIENT\ET\_HSM\_NETCLIENT\_SERVERLIST registry key.

### **b. Install ProtectToolkit C (PTKC) Runtime**

SafeNet PTKC is a cryptographic service provider using PKCS #11 API standard to provide the highest level of security when it interfaces with SafeNet access provider software for encryption, decryption, digital signature and permanent key storage.

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <ol style="list-style-type: none"> <li>1. Run the <b>PTKcprt.msi</b> file in the folder <i>ptkc_runtime</i> and follow the instructions as displayed on the wizard dialog box to complete the installation.</li> <li>2. When the setup complete window displays, click <b>Close</b> to exit. The driver's installation directory is automatically appended to PATH environment variable.</li> <li>3. Next, copy <b>jcprov.jar</b> file from &lt;HSM DRIVER_DIR&gt;\Protect Toolkit C RT (e.g. C:\Program Files\SafeNet\Protect Toolkit 5\Protect Toolkit C RT) to &lt;UAS_DIR&gt;\tomcat\webapps\am5\WEB-INF\lib\.</li> <li>4. Restart AccessMatrix UAS server.</li> </ol> <p><b>c. (OPTIONAL) Install ProtectToolkit C (PTKC) Software Development Kit (SDK)</b></p> <div data-bbox="256 613 1264 748"> <p><b>i</b> <b>Note:</b> Installation of this driver is optional. It is only required to install on the machine used to sign and deploy the FM. For normal AccessMatrix Server runtime, it is not required to install this component.</p> </div> <p>This toolkit provides the pkcs#11 interface and talks to the access provider, which in turn routes the request to the HSM.</p> <ol style="list-style-type: none"> <li>1. Run the <b>PTKcpsdk.msi</b> file in the <i>ptkc_sdk</i> folder and follow the instructions as displayed on the wizard dialog box to complete the installation.</li> <li>2. Select HSM (local adapter or remote server) as the Cryptoki Provider when prompted during the installation.</li> <li>3. Click <b>Close</b> to complete the installation. The driver's library and binary directories are automatically appended to PATH environment variable.</li> </ol> |
| Linux | <p>Next, retrieve the Linux client drivers from SafeNet ProtectServer HSM package version 5.7.0. Install the SafeNet ProtectServer Client drivers, Network HSM Access Provider and ProtectToolkit C in the directory server, in this case, the AccessMatrix UAS server to ensure communications between the AccessMatrix server and SafeNet ProtectServer HSM. The term &lt;UAS_DIR&gt; is used to refer to the installation directory of AccessMatrix UAS on Linux platform using Apache Tomcat.</p> <p><b>a. Network HSM Access Provider</b></p> <p>This package enables connection to one or multiple HSMs. It acts as a data abstraction layer for which you can add multiple front-end Application Program Interfaces (APIs).</p> <ol style="list-style-type: none"> <li>1. Install this driver by using the following command:<br/> <code>rpm -ivh &lt;nethsm_package&gt;.rpm</code><br/> Example: <code>rpm -ivh PTKnethsm-5.7.0-RC2.x86_64.rpm</code> </li> </ol> <p><b>b. ProtectToolkit C (PTKC) Runtime</b></p> <ol style="list-style-type: none"> <li>1. Install this driver by using the following command:<br/> <code>rpm -ivh &lt;cprt_package&gt;.rpm</code><br/> Example: <code>rpm -ivh PTKcprt-5.7.0-RC2.x86_64.rpm</code> </li> <li>2. After installation, run the following command:</li> </ol> <pre>In -s /opt/safenet/protecttoolkit5/ptk/lib/libjcprov.so &lt;UAS_DIR&gt;/tomcat/bin/libjcprov.so</pre>                                                                                                                                                                                                                                                                           |

3. Restart AccessMatrix UAS server.

### c. (OPTIONAL) ProtectToolkit C (PTKC) Software Development Kit (SDK)

**i** **Note:** Installation of this driver is optional. It is only required to install on the machine used to sign and deploy the FM. For normal AccessMatrix Server runtime, it is not required to install this component.

1. Install this driver by using the following command:

```
rpm -ivh <cpsdk_package>.rpm
```

Example: `rpm -ivh PTKcpsdk-5.7.0-RC2.x86_64.rpm`

Perform the following steps after installation:

1. Add the PATH and LD\_LIBRARY\_PATH environment variables with the SafeNet ProtectServer HSM directories. For example:

```
export PATH=$PATH:/opt/safenet/protecttoolkit5/ptk/bin)
```

```
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/opt/safenet/protecttoolkit5/ptk/lib)
```

2. Add the SafeNet NETCLIENT Server List by adding the following line:

```
export ET_HSM_NETCLIENT_SERVERLIST=<HSM_IP>:<HSM_PORT>
```

**i** **Note:** Substitute HSM\_IP and HSM\_PORT with the actual HSM IP address and port.

3. Save the changes.
4. Run the `hsmstate` command to verify whether HSM Access Provider is installed successfully. The response should be:

```
HSM device 0: HSM in NORMAL MODE. RESPONDING. Usage Level-0%
```

**!** **Important Note:** Perform the same steps above (i.e. setting of environment variables) if the OS user used to install the drivers is not the same with the OS user used to start the AccessMatrix server.

## Test the Connection to SafeNet ProtectSever Network HSM

To view the HSM status.

1. At the Windows command prompt, type the utility command `ctconf` and press Enter. Notice it returns the current adapter configurations that include the type of model, serial number and more.

## Uninstall SafeNet ProtectServer Network HSM

If you need to disconnect SafeNet ProtectServer Network HSM on AccessMatrix server, see the steps below to uninstall the program and drivers.

---

For Windows platform, launch **Windows Control Panel > Programs and Features > Uninstall a program**, and then double-click on the package to uninstall.

For Linux platform, you can remove the package(s) by using the program rpm, use the command `rpm -e, rpm -e <package name>; i.e. rpm -e PTKcpsdk`

| Package | Name     |
|---------|----------|
| Runtime | PTKcprt  |
| SDK     | PTKcpsdk |

---

## 3.2. Initialize Token and Change Pin (Windows OS)

A hard token represents the actual cryptographic device (e.g. the smart card). For soft token, it is a digital key that contains users' data encrypted in it. This section shows you how to initialize a token, set up user pin, security officer pin and more. A slot in the PKCS #11 model SafeNet uses represents a device interface. For example, a smart card reader would represent a slot. SafeNet ProtectToolkit-C supports three different slot types: user slots, smart card slots and admin slots. Refer to the SafeNet ProtectToolkit 5.7 Product Documentation, *ProtectToolkit-C Administration Guide* for more information.



**Note:** PKCS #11 denotes Public Key Cryptography Standard #11.

### Initialize Adapter

What is an adapter in the HSM? An adapter is the printed circuit board responsible for cryptographic processing in an HSM. If the Adapter is not initialized yet, perform the following steps.

1. Open the gCTAdmin utility from the SafeNet program folder. (e.g. *C:\Program Files\SafeNet\Protect Toolkit 5\Protect Toolkit C RT\gCTAdmin.bat*). Notice the **Safenet, Inc; Adapter Management** dialog box is displayed. On the **Enter Pin** dialog box, notice the last slot that labels, for example, "Admin Token (512345)" is automatically selected.
2. Select the **Security Officer** option and then enter its pin on the empty field below.
3. Click **OK** to initialize the adapter. Notice the adapter status is displayed in the **Adapter Management** dialog box.

### Create Slots

Before you initialize a token, check to see that a device interface (slot) for a token is available. If unavailable, you must create a slot before initialising a token.

1. Open gCTAdmin utility from the SafeNet program folder.
  2. Select **User**, enter the pin and click **OK** to log in.
  3. Select the **Token Configuration** icon or **Tokens...** command on the **Edit** menu. The **Manage Tokens** dialog box is displayed.
  4. Click the drop-down list of the slot to check if there is any (uninitialized token) slot. If no (uninitialized token) is available in the list, click **Done** to close this dialog box and proceed to create slot(s).
-

- 
5. On the **File** menu, select the **Create Slots...** command, or click the **Create Slots** icon.
  6. Enter the number of slots to be created.
  7. Click **OK**. The **Restart Alert** dialog box displays to restart CtAdmin. Click **OK**. Notice that the gCTAdmin utility restarts, and the number of (uninitialized token) slots are created. You can now initialize a token. Proceed to [Initialize Token](#).

## Initialize Token

1. Open gCTAdmin utility from the SafeNet program folder.
2. Select **User**, enter the user pin and click **OK** to log in.
3. After you log in successfully, select **Tokens...** on the **Edit** menu. The **Manage Tokens** dialog box is displayed.
4. Select an (uninitialized token) empty slot from the **Slot** drop-down list, and then select **Initialise**.
5. Enter the **Token Label**, **Security Officer PIN** and **User PIN**. Enter the number of the pin as listed on the dialog box for confirmation.
6. Click **Done**. The token is initialized.



**Note:** If no uninitialized token is available in the list, proceed to [Create Slots](#) before initializing a token.

## Change Token User PIN

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
2. After you log in successfully, select **Tokens...** on the **Edit** menu. The **Manage Tokens** dialog box is displayed.
3. Select the desired token for which the User PIN is to be changed.
4. Click **User PIN**.
5. Enter the **Current PIN**, **New PIN** and **Re-Enter the New PIN** to confirm.
6. Click **OK** to apply the change.

## Change Security Officer (SO) PIN

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
  2. After you log in successfully, select **Tokens...** on the **Edit** menu. The **Manage Tokens** dialog box is displayed.
-



- 
3. Select the desired token for which the SO PIN is to be changed.
  4. Click **SO PIN**.
  5. Enter the current **SO PIN**, **New PIN** and **Re-Enter the New PIN** to confirm.
  6. Click **OK** to apply the change.

## Reset/Reinitialize Token

1. Open the gCTAdmin utility from the SafeNet program folder and login as User.
2. Select the **Tokens...** on the **Edit** menu.
3. Select the desired token that needs to be reset and then click **Reset**.
4. Enter the current **SO PIN** and click **OK**.
5. Enter the **Token Label**, **Security Officer PIN** and **User PIN**.
6. Click **OK** to apply the changes. Notice the changes of the token label and its SO PIN and User PIN on the same slot.
7. Click **Done**.

## Remove Slots

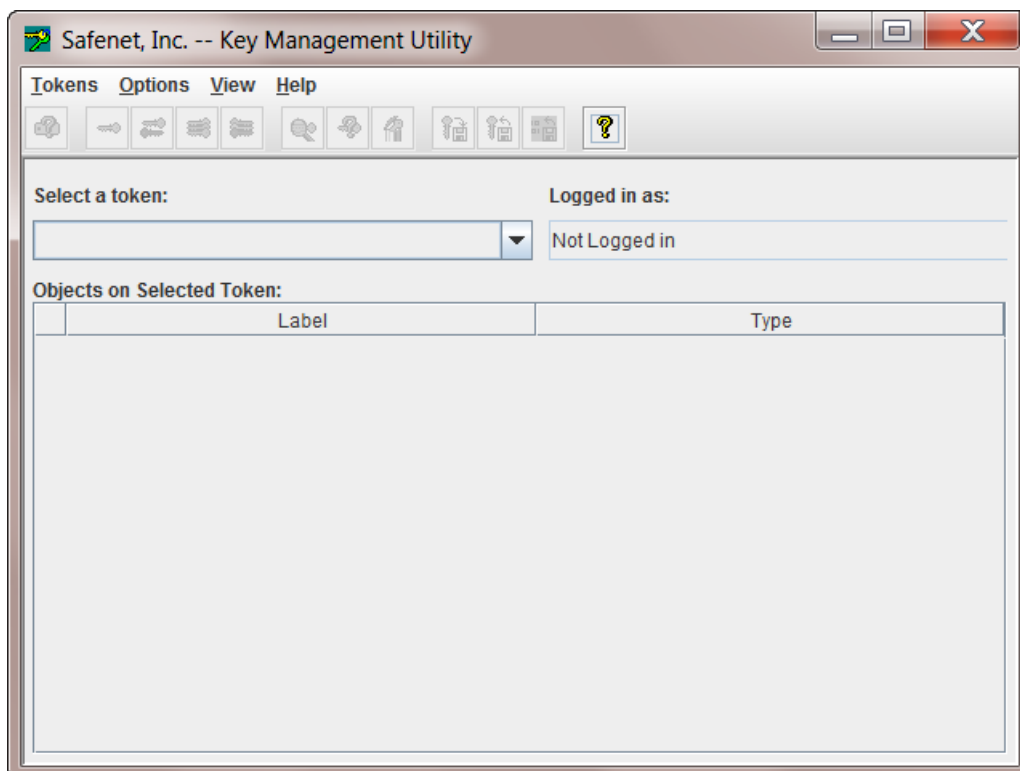
1. Open the gCTAdmin utility from the SafeNet program folder and login as User.
  2. Select **Delete Slots...** on the **File** menu. The **Delete Slots** dialog box is displayed.
  3. Select a slot to delete.
  4. Click **Delete**.
  5. Click **OK** to restart CtAdmin.
-

---

## 3.3. Manage Keys

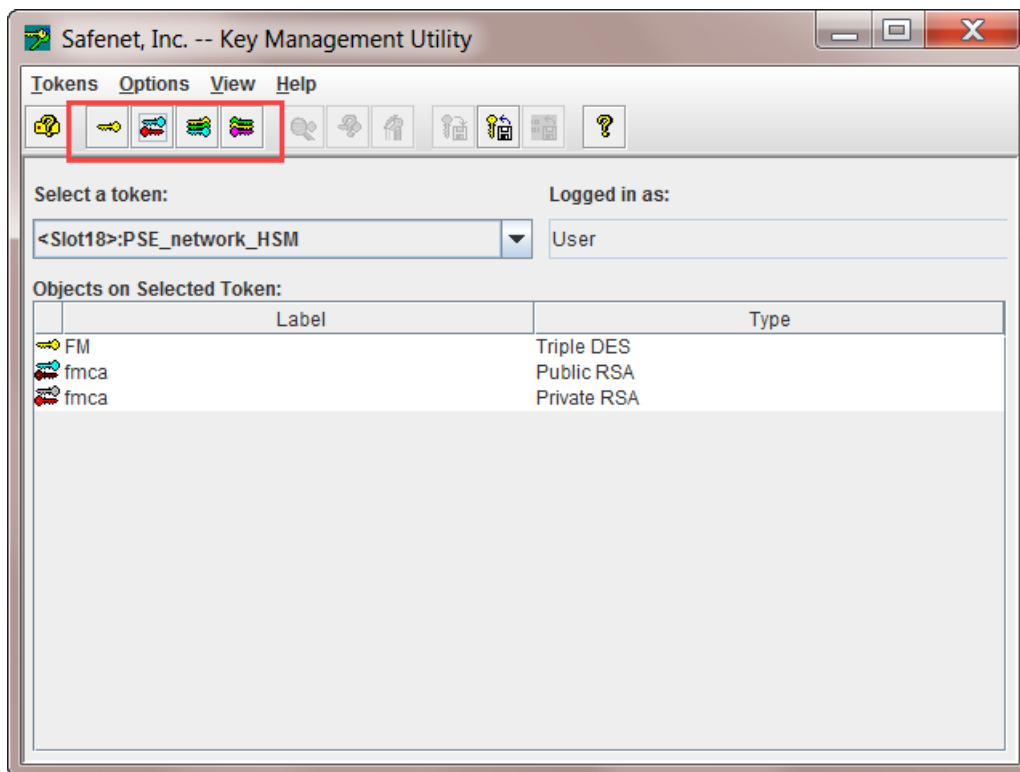
SafeNet ProtectServer HSMs provide high security by storing the cryptographic keys in the hardware. Use the Key Management Utility (KMU) tool in SafeNet Protect Toolkit C Runtime to manage keys. It provides a graphical user interface for key management functions using a PKCS #11 sub-system. KMU in SafeNet Protect Toolkit C Runtime release 5.7.0 is not compatible for use with SafeNet ProtectToolkit-C version 4.0 or less.

When KMU is opened, all toolbar functions are initially disabled. You must select a token from the **Select a token** drop-down list to start. It contains all the available tokens. The assigned label name displays initialized tokens. The uninitialized tokens are displayed as <Slotn>:<uninitialized token>. If there is no initialized token in this list, use gCTAdmin or the command-line utility `ctconf` to perform these operations. KMU is unable to initialize tokens and change pins.



### Generate Keys

Once a token has been selected, log in with the User pin number. The pin is authenticated, and a list of keys within the tokens are displayed in the **Objects on Selected Token** list to generate keys. See the screenshot below, the command icons on the toolbar are enabled for use to generate keys.



## Learn about Key Mechanisms and Key Attributes

The key exchange protocol is an essential part of a cryptographic mechanism to protect secure end-to-end communications. Below are the key mechanisms for cryptography you can find them in Key Management Utility (KMU) and the key attributes when you generate or create a secret key.

### Key Mechanisms

AccessMatrix server uses mainly the following key mechanisms. See below for its descriptions.

| Mechanism  | Description                                                                                                                                                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AES        | Advanced Encryption Standard (AES) is a symmetric block cipher used to protect classified information and is implemented in software and hardware to encrypt sensitive data. The cipher AES-256 is used among other places in SSL/TLS across the Internet.                                                                                                    |
| Triple DES | Triple Data Encryption Standard is a type of computerized cryptography where block cipher algorithms are applied three times to each data block. The key size is increased in Triple DES to ensure additional security through encryption capabilities. It is one of the commonly-used cryptography mechanism in the industry to prevent brute force attacks. |

| Mechanism | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RSA       | RSA algorithm is the basis of a cryptosystem - a suite of cryptographic algorithms that are used for specific security services or purposes which enables public-key encryption and is widely used to secure sensitive data, particularly when it is being sent over an insecure network such as the internet. Private Key is used for authentication and a symmetric key exchange during the establishment of an SSL/TLS session. It is a part of the public key infrastructure that is generally used in case of SSL certificates. In KMU, you can find the RSA mechanism in the <b>Generate Key Pair</b> dialog box. |

## Key Attributes

| Attribute Name | Description                                                                                                                                                                                                                                                                                                                                             |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Persistent     | Indicates whether a key object has been created for all the sessions. If you disable it, the key is only visible for the current session and will be destroyed when the session ends.                                                                                                                                                                   |
| Private        | Indicates whether users need to authenticate to the key's HSM token before they can access the key object. Defines whether the user PIN protects the object. A private object is only accessible to an application that has supplied the user PIN.                                                                                                      |
| Sensitive      | Indicates whether the key object can be extracted from the Hardware Security Module (HSM) in clear. The key object includes the values of all key attributes. In other words, if a key is <b>Sensitive</b> , the key's value will not be revealed in plain text/clear. Once a key becomes <b>Sensitive</b> , it cannot be modified to be non-sensitive. |
| Modifiable     | Indicates whether a key object can be changed after creation. Changing the key object involves changing the object's attributes.                                                                                                                                                                                                                        |
| Wrap           | Indicates whether the key can encrypt other keys that are in the HSM.                                                                                                                                                                                                                                                                                   |
| Unwrap         | Indicates whether the key can decrypt encrypted key material that is in the HSM.                                                                                                                                                                                                                                                                        |
| Extractable    | Indicates whether the key can be extracted from the HSM in encrypted form. Any user of the HSM can control the key encrypting key. It is recommended to use the <b>Exportable</b> rather than the <b>Extractable</b> property.                                                                                                                          |
| Export         | This attribute is similar to the <b>Wrap</b> attribute in that it specifies that the key may be used to encrypt a second key so that it may be extracted from the HSM in an encrypted form. Unlike the <b>Wrap</b> attribute, however, only the security officer may specify this attribute.                                                            |

| Attribute Name | Description                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exportable     | Indicates whether the key can be exported from the HSM in encrypted form. However, the key encrypting key (KEK) needs to be controlled by a security officer, not by a standard user. |
| Import         | This attribute is similar to the <b>Unwrap</b> attribute. It is used to determine if a given key can be used to unwrap encrypted key material.                                        |
| Derive         | Indicates whether other keys can be derived from the key.                                                                                                                             |
| Encrypt        | Indicates whether the key can be used to encrypt data.                                                                                                                                |
| Decrypt        | Indicates whether the key can be used to decrypt data.                                                                                                                                |
| Sign           | Indicates whether the key can be used to generate digital signatures or message authentication codes (MACs).                                                                          |
| Verify         | Indicates whether the key can be used to verify digital signatures or message authentication codes (MACs).                                                                            |

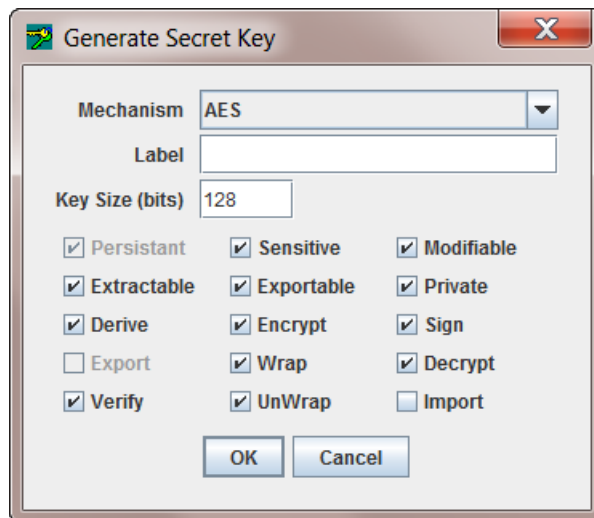
## Create Secret Key

A secret key is a cryptographic key or data that carries information used to encrypt and decrypt messages that are only known to the parties involved in a secured communication. The shared secrets can be OTP, pass phases, a random number or others.

Prerequisites before you create a secret key:

- A new slot has been created.
  - A new token has been created and initialized. refer to [Initialize Token](#) for more information.
1. Open the KMU tool and select the token that needs a new secret key. The **Enter PIN** dialog box is displayed.
  2. Log in as a **User** for the token.
  3. From the Option menu, select **Create > Secret Key**. The **Generate Secret Key** dialog box is displayed.
  4. Enter the key's label and select the mechanism. Triple DES mechanism is recommended.

- 
5. Check the options to select the required key attributes. You must discuss with your client(s) about the required security level to make these selections.



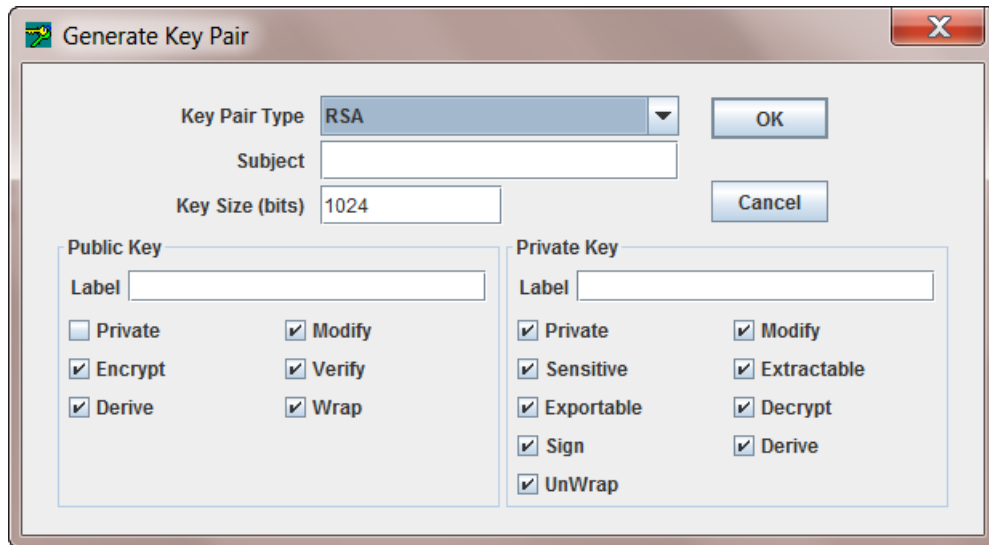
6. Click **OK**. A secret key is created.

## Create Key Pair

Key Pair is a pair of associated cryptographic keys that consists of a private key and a public key. Public/private key pairs are used by Crypto API to encrypt or sign data asymmetrically. Asymmetric encryption is used mainly to encrypt and decrypt session keys and digital signatures.

1. Open KMU tool and select the token that needs a key pair. The **Enter PIN** dialog box is displayed.
  2. Log in as a User for the token.
-

- From the **Options** menu, select **Create > Key Pair**. The **Generate Key Pair** dialog box is displayed.



- Select the **Key Pair** type.
- Enter the subject and the key labels for both the public and private keys.
- Select the required key attributes.
- Click **OK** to generate key pair.

## Delete Key

You can have many keys on a token in the list. See the following steps to remove one or more keys. For key pair, deleting one public key will not remove the connected private key.

- Open KMU tool and select the token within which the key resides. The **Enter PIN** dialog box is displayed.
- Log in as a User for the token.
- Right-click the key in the **Objects on Selected Token** list, and then select **Delete Key** on the pop-up menu. Alternatively, select **Options > Delete** from the menu bar.



### Notes:

- During the removal of key(s), no alert box prompts for confirmation.
- Use <CTRL> <Click> to randomly select more than one key, or use <Shift><Click> to select more than one key for deletion in the list.

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## Verify Key Check Value (KCV)

You can check that a key matches an expected key-value, without revealing anything about the actual key value, by viewing its Key Check Value (KCV). The KCV is a standard technique for obtaining an identification fingerprint from a key. The mechanism used, compatible with AS 2805, is simply the first three hex digits obtained by encrypting.

1. Open KMU tool and select the token within which the key resides. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. Select a key from the **Objects on Selected Token** list and click the **View** button on the toolbar. Alternatively, select **Options > View** from the menu bar. The **View KCV** dialog box is displayed.

## Set SafeNet Security Mode for Multiple Custodians Key

If you are exporting a key using the standard multiple custodians method, set the **Weak Mechanisms** flag to ON to generate key components of a transport master key or a key-encryption key (KEK). The multiple custodians method is where a key is split into shares and distributed among multiple custodians.



**Note:** Refer to SafeNet PDF document, *PTK-C Administration Guide* for more information about the security flag options.

1. Open gCTAdmin utility from the SafeNet program folder. (e.g. *C:\Program Files\SafeNet\Protect Toolkit 5\Protect Toolkit C RT\gCTAdmin.bat*).
2. Enter the **User PIN** and press **Enter** to log in.
3. Select **Edit > Security Mode** on the main menu. The **Manage Tokens** dialog box is displayed.
4. Select the **Custom** radio button to display the **Custom Security Flags** dialog box.
5. Click the **Weak PKCS#11 Mechanisms** checkbox and click **OK** to close the dialog box.
6. Click **OK** in the **Modify Security Mode** dialog box. A restart alert message is displayed.
7. Click **OK** to restart CtAdmin. You can now proceed to create the key components. The key components can be created by using either the **Generate Key Components** or the **Enter Key from Components** function of the KMU tool.



**Important Note:** This security flag is not necessary when using the **Single Custodian** method where the key is used to encrypt a key or a set of keys

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to be exported. Also, by setting this flag ON, it compromises security. After generating the key components, repeat the steps above and then turn OFF the **Weak PKCS#11 Mechanisms** flag.

## Create Key by Generating Key Components

**i** **Note:** Before you create a key by generating the key components, make sure you have performed the steps outlined in [Set SafeNet Security Mode for Multiple Custodians Key](#).

1. Open the KMU tool and select the token that needs a new key. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. From the **Option** menu, select **Create > Generate Key Components**. The **Create Key Components** dialog box is displayed.
4. Refer to [Learn about Key Mechanisms and Key Attributes](#) to select the required cryptography mechanism in the drop-down list.
5. Enter the label of the key.
6. Refer to [Learn about Key Mechanisms and Key Attributes](#) to check/uncheck the required key attributes in the list.
7. Click **OK** and enter the number of components needed to be created for the key. At least 2 components must be created.
8. Click **OK**. The **Information Message** dialog box is displayed.
9. Click **OK**. KMU generates the random numbers of the 1st key components based on the selected mechanism.
10. Click **OK** to generate the key component. Notice another **Information Message** prompts to generate the second key component.
11. Click **OK**. KMU generates the random numbers of the 2nd key components based on the selected mechanism, and so on, depending on the number of components you entered during the previous step.
12. Click **OK** to generate the key component. A secret key is created.

## Create Key by Entering Key Components

**i** **Note:** Before you create a key by entering the key components, make sure you have Set SafeNet Security Mode for Multiple Custodians Key.

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1. Open the KMU tool and select the token that needs a new key. The **Enter PIN** dialog box is displayed.
  2. Log in as a User for the token.
  3. From the **Option** menu, select **Create > Enter Key** from Components. The **Enter Key Components** dialog box is displayed.
  4. Refer to [Learn about Key Mechanisms and Key Attributes](#) to select the required cryptography mechanism in the drop-down list.
  5. Enter the label of the key.
  6. Refer to [Learn about Key Mechanisms and Key Attributes](#) to check/uncheck the required key attributes in the list.
  7. Click **OK** and enter the number of components needed to be created for the key. At least 2 components must be created.
  8. Click **OK**. The **Information Message** dialog box is displayed.
  9. Click **OK**. KMU asks for the 1st key component value (hex code). Fill in the component value; the KCV value will be generated automatically. Check the matching KCV with the component value.
  10. Click **OK** to generate the key component. KMU tool asks for the 2nd key component value (hex code).
  11. Paste the key component into the field, and check the matching KCV with the component value. The KCV has generated automatically. Depending on the number of components you entered during the previous step, KMU asks for the following key component value if more.
  12. Click **OK** to generate the key component. A secret key is created.

## Generate HSM Database Storage Key

Use HSM Data Storage key to prevent data loss. SafeNet provides data loss or failure-proof mechanisms. You create a database storage key based on the type of mechanism and the key attributes you select for its cryptographic security. Use either one of the methods above to generate HSM Database Storage Key. For more information, refer to the section in [UAS OneSpan FM with SafeNet ProtectServer HSM Implementation Guide - Generate HSM Data Storage Key](#).

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## Generate HSM Transport Master Key

It is necessary to generate a transport key to be available for transfer them from the generation site to the location where it will be used. Transport key should be achieved securely by SafeNet Key management techniques. Use SafeNet Key Management Utility (KMU) to generate the transport master key securely. Use either one of the methods above to generate HSM Transport Master Key. For more information, refer to [UAS OneSpan FM with SafeNet ProtectServer HSM Implementation Guide - Generate HSM Transport Master Key](#).

## Generate HSM Key Encryption Key (KEK)

Use KMU to generate the key-encryption keys for the high-security purpose of securing an extensive database system or a storage network using the concept of key wrapping. The purpose of this key-encryption key (KEK) is to export the HSM DPX transport key. KEK is a special key that is created with the Wrap attribute that is usually created as split custodian keys because of their high-security nature. Same as the procedural steps as listed in [UAS OneSpan FM with HSM Configuration - Generate HSM Transport Master Key](#). Enter the key's label, select the mechanism and the key attributes accordingly.



### Note:

- As an HSM KEK is also created using the multiple custodians' method, turn on the **Weak Mechanisms** flag to generate the key components before creation.
- Specify the number of KEK components when prompted for the **Number of components to create**.

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## 3.4. Back up the Selected Keys to Smartcards

You must back up the secret keys for transferring to other machines, for off-site storage or subsequent recovery for protection. SafeNet ProtectToolkit - C allows this backup for later recovery to other machines. There are no fixed rules to which keys should be backed up. The individual key owners or the security officers decide which keys need to be backed up for protection. In general, keys that are hard to be recreated or reconstructed by any means should be backed up. These may include generated key values or long keys manually entered by multiple custodians.

If you are using the multiple custodians method, the key is split into multiple shares. Key splitting scheme selection allows you to backup/recover:

- All of the custodians using the standard scheme or,
- By specifying the minimum number of custodians with the N of M scheme. See the prerequisites below for more information about the N of M scheme.



### Notes:

- Not all keys can be backed up and encrypted to the storage mediums such as the files to be transferred to other machines, smartcards or to the screen directly, as specific key attributes must be set to allow backup. The setting of key attributes is, therefore, an important consideration when creating keys used for backup operations. For example, the backup method requires the key attribute, **Wrap**, to be set to "True" for the wrapping key to back up another key. It also requires the key attribute **Exportable** to be set to "True", and the wrapping key is then known as an export key.
- Wrapping and export key should not be used for decryption, take note of the private key for decryption. Otherwise, the extractable/exportable key will be decrypted directly.
- Both the GUI tool KMU and the command line tool `ctkm` can be used to create keys and change their attributes.
- In the N of M scheme, the single custodian is not required to recover the key because the key will be rejected if it is corrupted when import. The custodians will be prompted for another card to continue.

Before you export keys to the smartcards, check the following prerequisites.

- Ensure a valid key or the keyset can be backed up with the required key attributes.
  - A connected smartcard reader is connected to the HSM.
  - There is sufficient disk space in the smartcards to back up the keys.
-

- 
- Check if you have to export the keys by using 1 custodian method or multiple-custodians method. Refer to the *SafeNet PTK-C\_Administration Guide* for more information if need.
  - Check if you will recover the key(s) using the key splitting N of M scheme method. If yes, define with your team or customer, the N and M values. This N of M scheme means that the key is split in such a way that the original key will be recovered with the co-operation of any of the custodians with a user-specified, a minimum number of custodians being required (N) of the total number of custodians (M). For example, if M = 3 and N = 2, only two out of the three custodians need to present their smart cards to recover the key.

**Notes:**

- Key N must be a minimum of 2, maximum equal to the total number of custodians specified M.
- N cannot be set equal to M.

## Export the Selected Keys to Smartcards (1 Custodian)

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. Select a key to back up in the **Objects on Selected Token** list. Use <CTRL> <Click> to randomly select more than one key, or <Shift><Click> to select all keys in the list if required.
4. From the **Options** menu, select **Export**. The **Export Key(s)** dialog box is displayed.
5. Select a wrapping key in the drop-down list.

| Method              | Wrapping Key                                                                                  |
|---------------------|-----------------------------------------------------------------------------------------------|
| Single custodian    | The desired wrapping key in the list used to encrypt the key (or set of keys) to be exported. |
| Multiple custodians | <Random key>                                                                                  |

6. Check the **Write to smart card(s)** option.
  7. Select the smart card slot and specify the batch name. The number of the custodians will be automatically set to 1 if the **Use N of M** option is not selected. Click **OK** to continue.
  8. KMU tool will ask for the **User Name** and **Smartcard PIN**. Specify the user name and smartcard PIN, and click **OK** to continue.
  9. Upon successful export, an **Export Successful** message is displayed.
-

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## Export the Selected Keys to Smartcards (Multiple Custodians)

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. From the **Options** menu, select **Export**. The **Export Key(s)** dialog box is displayed.
4. Do not select any wrapping key, and select the **Write to the smart card(s)** option and the **Use N of M** option. Specify the smartcard slot, batch name, number of custodians and number of custodians for recovery. The minimum number of custodians for this process is 3, and the minimum number of custodians for recovery is 2. Click the **OK** button to continue.
5. KMU tool will ask for the **User Name** and **Smartcard PIN**. Specify the user name and smartcard PIN, and then click **OK** to continue. Steps 4 and 5 will be repeated according to the number of custodians specified. Upon successful export, an **Export Successful** information message will be displayed.



**Note:** If you are exporting keys using the standard multiple custodians' method, you must set the **Weak Mechanisms** flag to ON. Refer to [Set SafeNet Security Mode for Multiple Custodians Key](#) for more information.

## Export the Selected Keys to a File

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. From the **Options** menu, select **Export**. The **Export Key(s)** dialog box is displayed.
4. Select the wrapping key and select the **Write to file** option.
5. Enter the path and filename of the file to be created in the **File to write** field. If a file with the same filename already exists at this location, then it will be overwritten. Alternatively, you can browse to a location and enter a filename by clicking on the ... button next to the **File to write** field.
6. Click **OK** to export to a file.

## Export the Selected Keys to the Screen

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
-

- 
2. Log in as a User for the token.
  3. From the **Options** menu, select **Export**. The **Export Key(s)** dialog box is displayed.
  4. Select a wrapping key and select the **Write encrypted part to the screen** option.
  5. Select **Single Part** or **Multi Part** export.
  6. Click **OK** to export to the screen.
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## 3.5. Import the Selected Keys from Smartcards

### Import the Selected Keys from Smartcards (1 Custodian)

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. From the **Options** menu, select **Import Key(s)**. The **Import Key(s)** dialog box is displayed.
4. Select the unwrap key and select the **Read from smart card** option. Select the smartcard slot in the drop-down list.
5. Click **OK** to continue.
6. Enter the smartcard PIN and click **OK** to import the key.

### Import the Selected Keys from Smartcards (Multiple Custodians)

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
2. Log in as a User for the token.
3. From the **Options** menu, select **Import Key(s)**. The **Import Key(s)** dialog box is displayed.
4. Select **Read from smart card(s)** option and do not specify the un-wrap key. Select the smartcard slot and click **OK** to continue.
5. Enter the smartcard PIN and click **OK** to continue.
6. Remove the current smartcard and insert another smartcard when prompted to do so, and click **OK** to continue.
7. Steps 5 and 6 will be repeated according to the number of custodians required as specified during the export process. The key is imported after the minimum number of keys required are entered.

### Import the Selected Keys from a File

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
  2. Log in as a User for the token.
-



- 
3. From the **Options** menu, select **Import Key(s)**. The **Import Key(s)** dialog box is displayed.
  4. Select the unwrap key and select **Read from selected file** option.
  5. Next, specify the file location to read.
  6. Click **OK** to import the key.

## **Import encrypted parts to Key Management Utility**

1. Open KMU and select the token that needs to be backed-up. The **Enter PIN** dialog box is displayed.
  2. Log in as a User for the token.
  3. From the **Options** menu, select **Import Key(s)**. The **Import Key(s)** dialog box is displayed.
  4. From the **Unwrap Key** drop-down list, select the wrapping key that was used to create the backup.
  5. Next, select **Import encrypted parts**.
  6. Select either **Multi-Part** or **Single Part** as applicable and click **OK** to continue.
  7. Enter the encrypted key (or key parts) and click **OK** to import the key. If the import key operation is a success, the message "Import command succeeded" is displayed. The newly imported key is also displayed in the **Objects on Selected Token** list in the main Key Management Utility interface.
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## 3.6. Configure Secure Messaging System (SMS)

SafeNet ProtectServer Network HSM supports a secure messaging system (SMS). It communicates with applications such as Functionality Modules(FM) requests in encrypted form across the network. For more information, refer to SafeNet document, *Network HSM Installation and Configuration Guide - Chapter 3: Secure Messaging System (SMS)*.

### Change Secure Messaging Mode

On Windows operating systems, user and system configuration information is stored in the Registry.

- **Temporary:** Temporary configuration changes are made using environment variables. Since environment variables are not hierarchical, the hierarchy is implicitly defined by the name of the variable.
- **User:** User configuration changes are made in the registry tree starting from HKEY\_CURRENT\_USER\SOFTWARE\SafeNet.
- **System:** System configuration changes are made in the registry tree starting from HKEY\_LOCAL\_MACHINE\SOFTWARE\SafeNet.

The HSM supports two SMS modes, ADH (Anonymous Diffie Hellman) and HIMK (Host Interface Master Key). The default mode of the secure messaging system is ADH.

Perform the following steps to change to HIMK mode:

1. Create the following attribute and set the value to HIMK.

`ET_PTKC_SMS_MODE = HIMK`

2. To change back to ADH, set the above attribute to ADH.

In order to allow the client to use the HIMK as well, above setting needs to be applied to the client machine. In addition, the client machine also needs to create following attributes:

`ET_PTKC_<serial>_HIMKLABEL = <HIMK label>`  
`ET_PTKC_<serial>_HIMKVALUE = <HIMK key value>`

Refer to *SafeNet PTK-C Administration Guide - Messaging Mode Configuration* for more information.

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## Configure Messaging Mode

Refer to *SafeNet PTK-C Administration Guide - Secure Messaging and Configuring Session Protection* for the configurations.

The ADH can be configured in two different modes, i.e. by the number of blocks or by the number of hours.

To configure the ADH, the following attributes should be set accordingly:

- By the preset number of blocks trigger value (default value is "2<sup>23</sup>").

```
ET_PTKC_SMS_BLOCKS = 10000
```

- By preset number of hours trigger value (default value is "24").

```
ET_PTKC_SMS_HOURS = 4
```

## Configure Session Protection

Refer to *SafeNet PTK-C Administration Guide - Configuring Session Protection* for details.

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## 3.7. Setup Functionality Module (FM) in HSM

Functionality Module (FM) is a segment of custom program code operating inside the HSM to provide additional or changed functionality of the hardware. i-Sprint developed E2EEFM and others. Each functionality module is distributed with a certificate so its identity can be verified. This certificate needs to be placed in the admin token by the administrator.

### Load a Functionality Module

The administrator can use the functionality module management utility `ctfm` for the SafeNet ProtectToolkit-C environment to manage the functionality modules on ProtectServer HSMs.

To install FM to the HSM, run the `ctfm` utility with following commands in the command line:

```
ctfm i [-a<device>|-A] -c<certFile> -l<certLabel> -<fmFile>
```

The `ctfm` utility will load a new functionality module into the specified HSM device or load a new FM into all HSMs by specifying the `-A` tag. The device administrator's PIN is required.

For more information about the command syntax for the `ctfm` utility, type `ctfm -h` in the command line.

### Query Functionality Module Status

To query the functionality module status, run the `ctfm` utility with the following commands in the command line:

```
ctfm q [-a<device>|-A]
```

The `ctfm` utility queries specified HSM device or query all HSMs by specifying the `-A` tag.

### Verify Functionality Module Signature

To verify the functionality module signature, run the `ctfm` utility with the following commands in the command line:

```
ctfm v [-a<device>|-A] -c<certFile> -l<certLabel> -<fmFile>
```

The `ctfm` utility verifies the specified HSM device or verifies all HSMs by specifying `-A` tag. The utility verifies if the functionality module has been signed without downloading it.

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## Disable Functionality Module

To disable the functionality module, run the `ctfm` utility with the following commands in the command line:

```
ctfm d [-a<device>|-A]
```

The `ctfm` utility will disable the specified HSM device or disable all HSMs by specifying `-A` tag.

---

## 3.8. Manage System Events Log

The event logs are used as a means of tracking serious hardware or operational faults, tamper events, and self-test error information. Event log entries are created when FMs are loaded or disabled. Each time a self-test fails, an unexpected event occurs at run-time, or a tamper occurs, information about the event is recorded to the event log. There can be up to 1024 events in the event log. Event records are written sequentially and labelled chronologically. If the date and time of a later entry are stating earlier than the entry preceding it, the real-time clock or audit information has likely been altered.

### View Event Logs

To view the event logs, run the `ctconf` utility with the following command:

```
ctconf -e
```

The administrator's pin is required to view event logs. Type the administrator's pin for this HSM and enter to view.

### Purge Event Logs

To purge event logs, run the `ctconf` utility with the following command:

```
ctconf -p
```

The administrator's pin is required to purge the event logs. Type the administrator's pin for this HSM and press Enter to purge.



**Note:** The event log will not be purged until the event log is full.

### List of Audited Events

Refer to [B. Event Log Error Types](#) for detailed information.

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## 3.9. Configure HSM

This section describes the HSM settings that should be configured, if necessary.

### Setup Security Mode

It is important for the administrator to determine how he can set the security mode based on the security policies and user roles. The steps below show you how to set up the security mode. Read about the security policies and user roles in the SafeNet PTK-C Administration Guide for more information.

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
2. Select the **Security Mode...** option from the **Edit** menu.
3. Perform either of the following:
  - a. Select one of the security mode options.
  - b. Alternatively, select **Custom**, select the required security attribute and then click **OK** to apply the selected custom security flag.
4. Click **OK** to modify the security mode.

### Setup Transport Mode



**Note:** Refer to the topic about "Using Transport Mode to Avoid a Board Removal Tamper" in the *SafeNet PTK-C Administration Guide* for detailed information on the transport mode when you need to remove the HSM hardware from the host system PCI bus. This is restricted to the administrator who can set the transport mode on the HSM.

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
2. Select the **Transport Mode...** option from the **Edit** menu.
3. Select the Transport Mode.
4. Click **OK** to apply the selection.

### Correct Clock Drift

The HSM hardware's internal clock may occasionally need to be adjusted, due to clock drifts and other timing differences between the HSM and the host system. The clock can be adjusted manually or synchronized with the host system's clock (recommended).

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
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2. Select the **Clock...** option from the **Edit** menu.
  3. Click **Synch** to synchronise the clock.
  4. Click **OK** button to apply the changes.
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## 3.10. Update and Tamper Firmware

This section describes the steps to update and tamper the HSM firmware.

### Update Firmware

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
2. Select **Upgrade Firmware...** option from the File menu.
3. Specify the upgrade file location.
4. Click **Open** to upgrade the selected firmware. The gCtAdmin utility restarts after the adapter firmware is successfully upgraded.

### Tamper Firmware

1. Open gCTAdmin utility from the SafeNet program folder and login as User.
2. Select **Tamper Adapter** option from the **File** menu.



**Important Note:** Tampering the adapter will erase all the keys and configuration data on this adapter. Do not continue if you are not a security officer or administrator to carry out this task. Click **Cancel** to discontinue.

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## 4. Appendices

### A. Security Mode Flag

Refer to *PTK-C Administration Guide - Security Flag Descriptions* for detailed information on the available flags and how they affect your installation and deployment.

### B. Event Log Error Types

| Name                       | Description                                                                      |
|----------------------------|----------------------------------------------------------------------------------|
| POST_ERR_SRAM_WRITE        | POST Error: Cannot write to SRAM                                                 |
| POST_ERR_SRAM_READ         | POST Error: Cannot read from SRAM                                                |
| POST_ERR_SDRAM_DATA_STUCK  | POST Error: SDRAM, bit stuck                                                     |
| POST_ERR_SDRAM_DATA_SHORT  | POST Error: SDRAM data bits short<br>Param 1. Bit number<br>Param 2. Value       |
| POST_ERR_SDRAM_ADDR_STUCK  | POST Error: SDRAM address bit stuck                                              |
| POST_ERR_SDRAM_ADDR_SHORT  | POST Error: SDRAM address bits short<br>Param 1. Bit number                      |
| POST_ERR_SDRAM_BAD_BYTESEL | POST Error: SDRAM bad bytes select                                               |
| POST_ERR_BAD_SECTOR0       | POST Error: POST Sector checksum is not correct                                  |
| POST_ERR_NOMEM             | Cannot allocate memory                                                           |
| POST_ERR_OS_HASH           | The OS hash value is incorrect                                                   |
| POST_ERR_KAT               | Known answer test failed<br>Param 1. Algorithm Identifier<br>Param 2. Error Code |
| POST_ERR_RNG               | RNG did not pass chi-squared test                                                |
| POST_ERR_NO_THREAD         | Unable to start POST Thread                                                      |
| POST_ERR_SMFS              | Secure memory file system error<br>Param 1. Error Number                         |
| POST_ERR_RTC               | Unable to access RTC                                                             |
| POST_ERR_SER               | Unable to access UART                                                            |

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| Name                  | Description                                                                            |
|-----------------------|----------------------------------------------------------------------------------------|
| EXCEPT_UNDEF          | An undefined instruction has been executed<br>Param 1. Address<br>Param 2. Instruction |
| EXCEPT_SWI            | A software interrupt generated<br>Param 1. Address<br>Param 2. Instruction             |
| EXCEPT_PREFETCH       | A Prefetch abort generated<br>Param 1. Address                                         |
| EXCEPT_DATA           | A Data abort generated<br>Param 1. Address                                             |
| EXCEPT_IRQ            | An unhandled IRQ received<br>Param 1. Identifier                                       |
| ERR_HOT_TAMPER        | Hot tamper detected                                                                    |
| LOG_FIRST_ENTRY       | Initial event entry                                                                    |
| LOG_INITIALIZING_SRAM | Initializing the SRAM after a tamper                                                   |
| LOG_EVENT_LOG_PURGED  | Event log has been purged                                                              |
| ERROR_ASSERT          | Runtime Assertion<br>Param 1. File<br>Param 2. Line                                    |
| ERROR_INIT_RESOURCE   | Out of resources in an initialisation<br>Param 1. File<br>Param 2. Line                |
| ERROR_INIT_PLATFORM   | Failed to detect hardware platform<br>Param 1. File<br>Param 2. Line                   |
| HEAP_INVALID_ADDRESS  | Heap Invalid block address<br>Param 1. Heap number<br>Param 2. Address                 |
| HEAP_MEM_FREED_TWICE  | Heap: Memory Freed twice<br>Param 1. Address                                           |
| PCCISES_TIMEOUT       | PCCISES: Timeout error on device<br>Param 1. Error                                     |

| Name                    | Description                                                                    |
|-------------------------|--------------------------------------------------------------------------------|
| PCCISES_BAD_STAT        | PCCISES: Bad device status<br>Param 1. Status                                  |
| PCCISES_BAD_DATA        | PCCISES: Bad input data                                                        |
| PCCISES_RNG_STUCK       | PCCISES: Continuous RNG test error<br>Param 1. Value                           |
| PCCISES_LNAU_EXCEPTION  | PCCISES: Large Number Arith Hardware exception (Unit,0)                        |
| PCCISES_FAILED_RESET    | PCCISES: Failed to reset                                                       |
| PCCISES_RESOURCES       | PCCISES: Insufficient resources to start driver                                |
| CPROV_OS_UPGRADED       | OS Upgrade performed<br>Param 1. Mod<br>Param 2. Version                       |
| CPROV_OS_UPGRADE_FAILED | OS Upgrade failed                                                              |
| PROT_NO_SMPR            | PROTECTION: HSM SMPR not found                                                 |
| PROT_CIPHER_ERROR       | PROTECTION: Cipher operation failed                                            |
| KEYGEN_ERR_PAIRWISE     | Key generation: Pair-wise consistency failure                                  |
| FM_OP_DOWNLOAD          | FM Download Performed<br>Param 1. Mod<br>Param 2. Version                      |
| FM_OP_DISABLE           | FM Disabled<br>Param 1. Mod<br>Param 2. Version                                |
| FM_MODULE_FAILED        | FM failed to load<br>Param 1. Mod<br>Param 2. Version                          |
| PTKC_CFG_CHNG           | SafeNet ProtectToolkit-C config change<br>Param 1. New Val<br>Param 2. Old Val |

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